

Douglas Gordon Daly

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Principal AI Engineer / AI Solutions Architect

SUMMARY

- Principal AI Engineer and technical leader with 20+ years building enterprise decision systems, operational intelligence platforms, and production AI solutions.
- Hands-on experience with RAG, agentic AI, tool orchestration, graph analytics, BigQuery, GCP, AWS, and Python.
- Deep background in observability, reliability, AI governance, and cross-functional technical leadership across utilities, financial services, media, retail, telecommunications, and aerospace/defense.

CORE EXPERTISE

AI Architecture | Production AI Systems | Agentic AI | RAG | Enterprise Integration | Operational Intelligence | Observability | MLOps | Graph Analytics

TECHNOLOGIES

Python | SQL | BigQuery | GCP | AWS | Gemini | LangGraph | MCP | SageMaker | Terraform | S3 | Neptune | GraphStorm | Bedrock | Kafka | Elastic Stack | Splunk

SELECTED EXPERIENCE

Principal Consultant | Daly Engineers LLC | Jan 2026 - Present

- Led analytical readiness assessment for Oracle/EBS and Maximo data migrated to BigQuery, establishing cross-domain enterprise data integration for construction operations and financial analytics at Consolidated Edison.
- Developed automated data profiling, type inference, duplicate detection, candidate key analysis, and cross-system join validation frameworks across 100+ source tables with millions of operational and financial records.
- Created a phased analytical maturity roadmap adopted as the foundation for anomaly detection, construction analytics, and AI-enabled decision support initiatives.
- Architected and implemented a governed analytics assistant leveraging Gemini, BigQuery, and tool-calling architectures to enforce analyst rules and operational controls with predictable, auditable execution.

Principal Consultant | Apex Systems / NBCUniversal | Jan 2026 - Present

- Built AWS-based AI and graph analytics pipelines using SageMaker, Terraform, S3, and containerized workloads to support NBCUniversal's digital twin initiatives.
- Developed graph-ready service dependency datasets from API telemetry and operational metrics, enabling graph-based dependency modeling and operational intelligence.

AI Instructor | SimpliLearn & Interview Kickstart | 2025 - Present

- Delivered professional training on LLMs, RAG, agentic AI, MCP, diffusion models, CLIP, and modern generative AI architectures to software engineers and technical professionals.
- Developed hands-on demonstrations, Jupyter notebooks, and end-to-end applications covering modern AI workflows, evaluation techniques, and production deployment considerations.

Data Scientist | Meta | Oct 2021 - Jan 2025

- Developed reliability measurement and forecasting frameworks for Meta FinTech, defining leading indicators and decision-support models that translated engineering initiatives into projected reliability outcomes for planning, prioritization, and performance management.
- Built attribution frameworks linking incidents, alerts, defects, and operational metrics to engineering organizations and service ownership, enabling accurate accountability and observability.
- Designed and developed the ReMon framework, enabling engineering teams to build comprehensive monitoring and alerting systems without requiring expertise in complex internal tooling; adopted across Meta FinTech and contributed to automatic detection of over 15 SEVs in 2024.

Data Scientist | Samsung Electronics | Dec 2020 - Oct 2021

- Developed gradient-boosting propensity models and automated audience-segmentation workflows for digital marketing campaigns, increasing conversions by 146% and campaign-attributed revenue by 229%.

Head of Data Science | Facteus | Oct 2017 - Nov 2020

- Led development of market-intelligence products on a 30B+ transaction dataset, transforming raw payment-card activity into merchant, industry, and consumer-behavior intelligence used for competitive analysis and strategic decision-making.
- Built large-scale entity-resolution systems that transformed billions of cryptic payment-card transactions into structured merchant, brand, industry, and storefront intelligence, enabling analytics across 2,500+ merchants representing more than 60% of observed card transactions and achieving over 99% transaction coverage.

Senior Manager, Operations Analysis | Capital One | Jan 2012 - Dec 2016

- Developed predictive capacity-management models forecasting peak-demand scenarios for customer-facing banking platforms, supporting infrastructure planning, resiliency investments, and operational risk management.
- Identified a critical observability gap in which major incidents were frequently detected through customer call centers rather than monitoring systems. Conceived, designed, and led adoption of RITHM, an operational intelligence platform that aggregated monitoring signals by business capability and ownership, reducing major incident detection times from hours to minutes.
- Secured executive sponsorship and partnered with CTO-level leadership to modernize enterprise monitoring strategy, expanding Splunk prototypes into Kafka and Elastic Stack platforms adopted across the organization. Recognized with the Capital One Business Innovation Award for contributions to operational intelligence and monitoring transformation.

Senior Principal Systems Engineer / Program Manager | Raytheon | Mar 2005 - Dec 2011

- Led systems engineering teams of 40-50 engineers specializing in modeling, simulation, and advanced aerospace systems, providing technical leadership and risk management for programs valued up to \$50M. Turned around an over-budget classified fixed-price technology program, delivering under budget and nearly doubling profit targets.
- Served as Principal Investigator for advanced radar resource-management systems, leading research, simulation, flight testing, and operational deployment across multiple aircraft programs. Developed patented fuzzy-logic scheduling technology deployed on major Department of Defense aircraft platforms.

ADDITIONAL EXPERIENCE

Data Scientist | Intuit | Apr 2025 - Jul 2025

SEO Solutions Architect | Nike | Nov 2016 - Sep 2017

SELECTED PROJECTS

GraphRAG Engine & LoRA Coverage Assistant

- Built a GraphRAG system for multi-hop benefits reasoning with traceable retrieval paths across entities, clauses, and supporting evidence.

MCP / Agentic Workflow Demos

- Developed hands-on MCP and agentic workflow demos illustrating tool orchestration, structured outputs, traceability, and evaluation patterns.

EDUCATION

MBA, UCLA Anderson | MS, Applied Statistics, Rochester Institute of Technology | MS, Electrical Engineering, University of Washington | BS, Physics, Humboldt State University

HONORS & AWARDS

- Lead inventor, U.S. Patent 8,635,622, Method and system for resource management using fuzzy logic timeline filling
- Capital One Business Technology Innovation Award